

Experimental Determination of the Spring Constant and Unknown Mass Using an Inertial Beam Balance

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Abstract

This study presents the experimental determination of the spring constant of an inertial beam balance and the measurement of an unknown mass using dynamic oscillations. The balance was subjected to a series of known masses, including the platform alone and additional loads ranging from 201 g to 1000 g. Each mass was displaced laterally and released, allowing free oscillations, which were recorded for 30 s using a smartphone camera. The platform's motion was tracked via a colored marker using the OpenCV Python library, and the displacement of the center of mass was extracted frame-by-frame. Fast Fourier Transform (FFT) analysis was performed on the position-time data to determine the dominant oscillation frequency for each mass. The oscillation periods were then used to calculate the spring constant through direct computation and linear regression of T^2 versus mass. Both methods produced consistent results within experimental uncertainty. Subsequently, the system was calibrated, and the unknown mass was determined using these established relationships with high accuracy. The k constant of the system is determined to be in the value 81 ± 2 N/m, and the unknown mass was determined within 1-8% accuracy.

Keywords: Inertial balance, FFT, spring constant determination

1 Introduction

Newton's first law of motions tells us that "An object at rest stays at rest and an object in motions stays in motion with the same speed and direction unless acted upon by an unbalanced force." This tells us about inertia which is the tendency to resist changes in the state of rest or motion. It is said that the mass of the object is said to be the measure of its inertia [1].

An inertial balance is a spring device that utilizes vibrations in order to change the velocity and record the inertia. Inertial mass is measured as a dynamic measurement, unlike gravitational mass which is instead measured as a static measurement [2]. This method is applied in orbiting platforms or space vehicles, where conventional gravitational methods of measuring mass are not possible. The inertial balance device is based on the principle of simple harmonic oscillation (SHO) where the time period of its oscillation T is given as [3]

$$T = 2\pi\sqrt{\frac{I}{k}} \quad (1)$$

where I is the moment of inertia and k is the restoring force constant from Hooke's law. The moment of inertia of the oscillating system (the platform and the added mass) is

$$I' = \frac{1}{3}m_s L^2 + mL^2 \quad (2)$$

where m_s is the mass of the lever arm spring, m is the added mass on the platform, and L is the distance of the clamp to the point of oscillation. By substituting I into our time period equation, we get

$$T = 2\pi\sqrt{\frac{\frac{1}{3}m_s L^2 + mL^2}{K}} \quad (3)$$

Since the platform mass m is much larger than the mass of the spring strips, the term involving m_s becomes negligible. Thus, the expression reduces to the familiar simple harmonic oscillator period, which

also implies that the square of the period is linearly proportional to mass

$$T = 2\pi\sqrt{\frac{m}{k}} \implies T^2 \propto m \quad (4)$$

or, if T_1 and T_2 are the time periods of oscillations of masses m_1 and m_2 respectively, then [3]

$$\frac{m_1}{m_2} = \frac{T_1^2}{T_2^2} \quad (5)$$

This equation show us that we can calibrate the inertial balance using known masses and their respective oscillations in order to determine the mass of an unknown object from its measured oscillation period [1].

This experiment aims to determine the spring constant k of the inertial balance device and experimentally determine the relationship between the oscillation period and mass of the system. Finally, after calibrating the system using known masses, we would use the oscillation period to determine the unknown mass of an object.

2 Methodology

An inertial beam balance was securely positioned near the edge of a rigid and level surface to minimize vibrational interference. A sequence of trial masses was employed, consisting of the platform's own mass m_0 and additional loads of 201 g, 289 g, 403 g, 502 g, 520 g, 799 g, 1000 g, and an unknown mass (actual mass is 150 g). Each mass was measured using a provided electronic weighing scale. For each measurement, the platform was displaced laterally by approximately two inches from its equilibrium position and released from rest, allowing it to undergo free oscillations about the equilibrium point.

To facilitate motion tracking, a circular colored marker (either red or green) was affixed to the periphery of the platform. The oscillatory motion was recorded for 30s per trial using a smartphone camera operating at a fixed frame rate and resolution. The resulting video sequences were processed frame by frame using the OpenCV (cv2) Python library. The marker was isolated through color-based thresholding in RGB space, and its centroid coordinates were extracted for each frame. From these coordinates, the instantaneous displacement of the platform's center relative to equilibrium was reconstructed as a function of time.

The time-domain displacement data were transformed into the frequency domain via a Fast Fourier Transform (FFT) to identify the dominant oscillation frequency f corresponding to each effective mass. Majority of the computation can be found in the provided Github repository. The oscillation period was then determined from the reciprocal of the dominant frequency, $T = 1/f$. The dependence of T^2 on the total oscillating mass m was analyzed to evaluate the effective spring constant of the system and to calibrate the inertial beam balance for subsequent determination of unknown masses.



Figure 1: An inertial balance. Note the thin spring and the platform where the mass rest [4]

3 Results and Discussion

Several methods were employed to determine the spring constant of the inertial balance system. In the first approach, the spring constant was calculated directly from individual measurements using

$$k = \frac{4\pi^2(m + m_0)}{T^2}, \quad (6)$$

where $m_0 = 57$ g represents the intrinsic mass of the platform. Multiple trials were conducted, and the resulting k values were averaged. The standard error of the mean was computed, yielding

$$k = 84.0 \pm 2.4 \text{ N/m}.$$

The second approach involved performing a linear regression on the measured T^2 versus m data, exploiting the relationship

$$T^2 = \frac{4\pi^2}{k}(m + m_0). \quad (7)$$

The slope of the best-fit line corresponds to $4\pi^2/k$, from which the spring constant was determined:

$$k = 78.41 \pm 2.8 \text{ N/m}.$$

These two values differ not so significantly, so we just put it in a ball park of 81 ± 2 N/m. Figure 2 illustrates the linear dependence of T^2 on m . Note that the 1 kg mass exhibits a slightly higher period relative to the general trend, likely due to increased air resistance associated with its larger size.

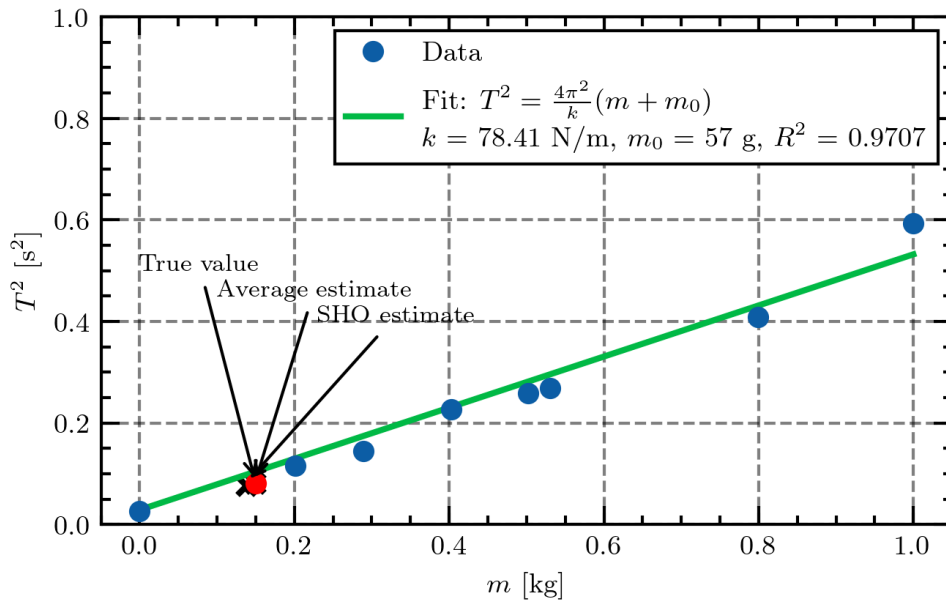


Figure 2: T^2 as a function of added mass m , demonstrating the expected linear relationship. Deviations at higher masses are attributed to increased air resistance. Note also the close agreement of the unknown value (red) against the calculated values

The unknown mass was determined using three complementary approaches. First, applying

$$m = \frac{kT^2}{4\pi^2}, \quad (8)$$

with k obtained from the averaging and linear regression methods. Second, the mass was estimated using the comparative equation (5). The results are summarized in Table 1.

Table 1: Determination of the unknown mass using different methods. Percent errors are calculated relative to the true mass of 150 g.

Method	Value [g]	Percent error [%]
Averaging	148.1	1.27
Linear regression	137.4	8.40
Mass comparison	147.4 ± 4.3	1.73

The results indicate a maximum percent error of 8%, which is within an acceptable range for this type of measurement. Consequently, the spring constant of the inertial balance was reliably determined, and the unknown mass was estimated with reasonable accuracy.

4 Conclusion

The experimental investigation successfully determined the spring constant of the inertial balance and measured an unknown mass with minimal error. Two complementary methods were employed: direct calculation from measured periods and linear regression of T^2 versus mass. The resulting spring constants, $k = 84.0 \pm 2.4$ N/m and $k = 78.41 \pm 2.8$ N/m, were in good agreement, demonstrating the reliability of the measurements. The unknown mass, estimated using these values and comparative techniques, yielded results within 8% of the true mass, confirming the accuracy of the method. The study highlights the effectiveness of using dynamic oscillations and FFT-based analysis to determine mass in systems where conventional gravitational methods are impractical. Overall, the inertial beam balance proved to be a reliable instrument for both calibrating spring constants and determining unknown masses, providing valuable experimental insight into simple harmonic oscillators and the relationship between oscillation period and mass in practical applications.

References

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Appendices

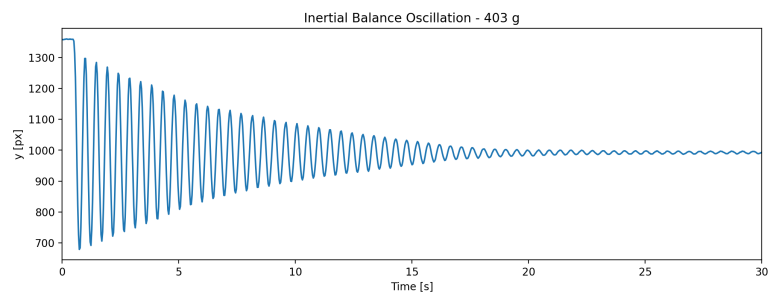
Table 2: Table of Contributors

Member	Contribution
Danielle Louis Dumas	Performed the experiment, wrote the methodology
Vercil Juan	Performed the experiment, wrote the results and discussion and abstract
Rwen Josh Narvarte	Performed the experiment, wrote the introduction and conclusion

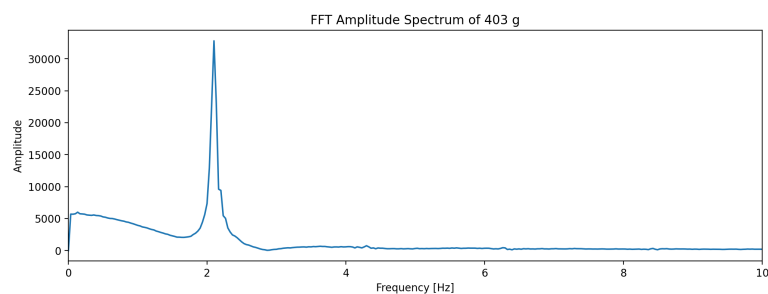
Documentation



Figure 3: Screen shot of the video taken during the experiment



(a) Time-domain of inertial balance oscillation



(b) Frequency domain of inertial balance oscillation

Figure 4: Frame by frame analysis of the videos taken led to the data of the oscillation, which was analyzed with FFT

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Code

All the code used for the graphing and statistical analysis for this report can be found in:

<https://github.com/silvernuke911/Physics157/tree/main/PHYSICS192%20CODE/E4B>